



SUN PROTECTION

Sunlight contains harmful ultraviolet rays that can damage skin cells. As a defensive barrier, skin contains special cells called melanocytes. These cells produce melanin, a natural brown pigment (coloring substance) that passes into neighboring epidermis cells. Here, melanocytes form a screen that blocks ultraviolet rays. In sunny weather melanin production increases, producing a suntan.

SKIN CELLS

Continually worn away, these dead, flat skin cells are replaced by cells that move to the surface from the lowest layer of the epidermis.

Receptors detect touch, cold, heat, or pain.

Flat, dead cells on surface of epidermis

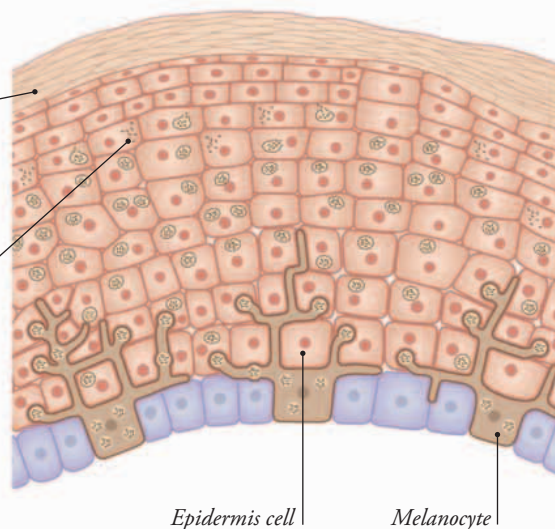
Melanin granules spread out in epidermis cells

This tiny muscle pulls the hair upright in cold conditions.

A hair follicle is a narrow pocket in the skin from which a hair grows.

The fat layer under the skin stores energy and helps keep you warm.

Sweat glands release watery sweat onto the skin's surface to cool you down.



Epidermis cell

Melanocyte

FINGERPRINTS

Packed with receptors, your fingers are incredibly sensitive. They also have tiny ridges that help you grip objects. On hard surfaces, such as glass, ridges leave behind patterns called fingerprints. There are three main types of patterns—loop, whorl, and arch—but each person's fingerprint is unique. That's why the police use them to identify criminals.



Loop

Whorl

Arch